COMPARISON OF TWO TRIPHASIC CONTRACEPTIVES WITH DIFFERENT PROGESTOGENS: EFFECTS ON METABOLISM AND COAGULATION PROTEINS

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Abstract

This study compared the effects of two triphasic oral contraceptives (OCs) taken for 6 pill cycles. One preparation contained levonorgestrel (EE/LN), the other a new progestagen, gestodene (SHG 415G). There were no effects on body weight, dietary habits, blood pressure, HDL-cholesterol or carbohydrate metabolism. Both OCs caused a small but statistically significant increase in plasma total cholesterol, LDL-cholesterol, LDL/HDL-cholesterol ratio and triglyceride concentration. Sex hormone binding globulin was increased by both preparations. Coagulation factor VII and fibrinogen were increased and antithrombin III levels reduced, indicating that both preparations had pro-coagulant activity. The gestodene triphasic preparation was associated with minor side effects similar to those experienced on EE/LN. The metabolic effects were similar despite the lower total steroid dose of the gestodene preparation.

Introduction

There is an association between combined oral contraceptive (OC) use and cardiovascular disease [1,2]. This may be related to the effect of the preparations on lipid metabolism, as there is a relationship between plasma lipoprotein levels and coronary heart disease (CHD) [3,4]. The risk of CHD is associated with a high plasma concentration of total and low density lipoprotein (LDL) cholesterol, reduced high density lipoprotein (HDL) cholesterol and, in some circumstances, high triglycerides (TG) [5].
Most studies have shown the increased risk of CHD associated with current [6] and past use [7] of OCs to be greatest in predisposed individuals. The problem was initially attributed to the oestrogenic component, and analysis of drug safety reports for several countries indicated a dose response trend [8,9].

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However, the progestagen content also appears relevant. Meade et al. [9] found a significant positive association between the dose of norethisterone acetate (NE) and deaths from stroke and CHD, and a possible excess of deaths from stroke and thromboembolism in women on OCs with a high levonorgestrel (LN) content. Some progestagens may increase blood pressure [9] and reduce glucose tolerance [10]. OCs can also have an effect on coagulation factors [11] and all these factors may be relevant in respect of vascular disease risk.
The effect of a progestagen depends on its relative biological activity and inherent oestrogenic and anti-oestrogenic/androgenic properties [12]. Androgenic and progestational steroids decrease HDL- cholesterol [13], the opposite effect to ethinylestradiol (EE) which tends to increase plasma triglycerides, VLDL, LDL and HDL cholesterol in a dose-dependent manner [14]. Triphasic oral contraceptives have been developed as a means of reducing the total steroid content of the pill, and to minimise unwanted metabolic effects. New progestagens have also been developed; one of these progestagens which acts more selectively on the pituitary/ovarian axis is gestodene. Gestodene is a derivative of 19-nor testosterone, and differs from levonorgestrel in having a double bond between C-15 and C-16. Gestodene (Δ-15 levonorgestrel) is approximately three times more potent than LN as a progestagen but has similar anti-oestrogenic/androgenic potency[15].
This study assesses the effects of a new and, as yet, unmarketed, triphasic preparation containing EE and gestodene (SHD 415G). The effects on lipoproteins, carbohydrate metabolism, sex hormone binding globulin and coagulation factors were compared with those of a relatively widely used triphasic OC containing the same amount of ethinylestradiol, but LN as the progestagen (EE/LN).

Methods

The patients included were healthy women aged 16 to 30 years who attended the Family Planning Clinic of the John Radcliffe Hospital requesting oral contraceptive therapy. Women were not recruited into the study if they had taken hormonal preparations in the previous 6 months or had a contra-indication to combined OC use. Other exclusion criteria were a body weight 25% above or below the ideal for their height, alcoholism, diabetes, a diastolic blood pressure of greater than 90mm Hg,a smoking habit of more than 10 cigarettes per day, a family history of hyperlipidaemia or the use of drugs which are known to affect lipid metabolism.
The study was performed in accordance with the Declaration of Helsinki, the women's consent was obtained and the study protocol was approved by the Central Oxford Ethics Committee. An initial assessment was made of each patient including age, family history, general health, obstetric and menstrual

history, diet, cigarette use and exercise. Any medication used was noted, and an attempt made to keep this constant throughout the study. Height, weight and blood pressure were measured, and a venous blood specimen was taken after a 12-hour overnight fast.
Patients were then randomly allocated to receive either EE/LN or SHD 415G. The details of the pills are shown in Table I. Over a 6-cycle period, each patient received one tablet of their allocated OC for the first 21 days of the 28-day cycle, followed by a 7-day 'tablet free' interval.

Table I Treatment Regimen

Menstrual Cycle:	Days 1 - 6	Days 7 - 11	Days 12-21	Days 21-28	Total Dose per cycle
EE/LN					
Ethinylestradiol (μg)	30	40	30	0	680
Levonorgestrel (μg)	50	75	125	0	1875
SHD 415G					
Ethinylestradiol (μg)	30	40	30	0	680
Gestodene (μg)	50	70	100	0	1650

The women were reviewed between days 15 and 21 of the first, third and sixth pill cycles. They were interviewed to ascertain changes in diet, medication and exercise; directly questioned about any adverse effects, and their weight and blood pressure were measured. Fasting venous blood samples were taken on each occasion for the measurement of plasma lipids, lipoprotein fractions, glucose, glycosylated haemoglobin, sex hormone binding globulin, coagulation factors VII, antithrombin III and fibrinogen.
The total cholesterol concentration and the cholesterol concentration in the lipoprotein fractions was measured enzymatically [16]. HDL-cholesterol was

measured after precipitation of LDL and very low density lipoproteins (VLDL) with heparin and manganese [17]. HDL subfractions in the supernatant were measured by precipitating HDL_2 with dextran sulphate and measuring the HDL_3 subfraction remaining in solution [18]. HDL_2 was then calculated by subtraction. The quantity of cholesterol in the various subfractions was then calculated by subtraction. Plasma triglyceride concentration was measured enzymatically using a Boehringer kit. Plasma glucose was measured enzymatically using glucose oxidase [19]. HbA1c was measured using thiobarbituric acid [20]. Sex hormone binding protein was measured by column chromatography [21].
Venous blood for analysis of coagulation factors was mixed with trisodium citrate 0.109M (9:1 v/v) and centrifuged immediately at 2,000g for 15 min to obtain platelet-poor plasma which was snap-frozen and stored in plastic tubes with liquid nitrogen.
Factor VII clotting activity was measured using a one-stage assay based on the prothrombin time using an artificially depleted substrate plasma [22].
Fibrinogen concentration was measured using the Clauss assay [23]. A commercial standard (Immuno) was used.
Antithrombin III (AT-III) activity was measured by an amidolytic microtitre assay based on the method described by Odegaard [24] using a specific synthetic tripeptide chromogenic substrate, S-2238 (Kabi Diagnostica, Stockholm), and British Standards obtained from the National Institute for Biological Standards and Controls.
The concentration of the various analytes was compared to baseline. Statistical analysis was performed using the analysis of variance (F test) to identify any significant treatment and time effects, and Duncan's multiple range test was also used where appropriate. The Chi-squared test was used to identify differences in reported side effects.

Results

Forty-three patients were entered into the study. Twenty-one were randomised to receive EE/LN, 22 to receive SHD 415G.
The two treatment groups were comparable at the start of the study with regard to age, weight, height, blood pressure, alcohol consumption, exercise, dietary habits and all biochemical/haematological measurements. Eleven women smoked, and although more smokers (7 women) were assigned to the EE/LN group, this was not statistically significant. Only one woman had ever been pregnant. Four patients on SHD 415G and 6 on EE/LN received other medication which was mainly antibiotics, antihistamines and vitamin supplements, and one woman in each group took a non-steroidal anti-inflammatory agent.

There were no statistically significant changes in blood pressure or body weight within the groups during the 6-cycle period and no difference between the two treatment groups (Table II). One patient in each group had a diastolic blood pressure of 90 mm Hg or systolic blood pressure of 150 mm Hg, but none had values exceeding this.

Table II Patient data
Results are means ± SEM

	SHD 415G				EE/LN			
	Pre-RX	Cycle 1	Cycle 3	Cycle 6	Pre-RX	Cycle 1	Cycle 2	Cycle 6
No. Patients	22	21	18	18	21	21	20	20
Age (years)	21.2 ± 0.7				20.4 ± 0.7			
Smokers (3-10/day)	4				7			
Weight (kg)	64 ± 2.6	64 ± 2.6	64 ± 2.5	64 ± 2.5	59 ± 2.3	60 ± 2.3	60 ± 2.3	61 ± 2.3
Blood Pressure (mmHg)								
Systolic	124 ± 2	124 ± 3	119 ± 3	122 ± 3	122 ± 3	118 ± 2	121 ± 2	119 ± 2
Diastolic	77 ± 1	76 ± 1	71 ± 2	75 ± 1	72 ± 2	72 ± 2	73 ± 2	72 ± 2
Changes of Concurrent Medication:		3	1	3		4	4	1
Changes in:								
Smoking habits		1	0	0		3	3	4
Eating habits		4	3	2		5	4	4
Drinking habits		1	2	1		1	3	2
Exercise		3	0	3		3	6	4

Differences between the groups are not statistically significant, and there is no significant time effect or treatment time interaction.

Table III Oral contraceptive effects on serum lipids

Results are means ± SEM in mmol/l

	SHD 415G					EE/LN				
	Pre-Tx	Cycle 1	Cycle 3	Cycle 6	Statist. Analysis	Pre-TX	Cycle 1	Cycle 3	Cycle 6	Statist. Analysis
Triglyceride	1.30 ± 0.11	1.37 ± 0.12	1.52 ± 0.10	1.60 ± 0.13	1	1.02 ± 0.09	1.22 ± 0.10	1.40 ± 0.14	1.17 ± 0.09	1*
Total chol.	3.49 ± 0.17	3.71 ± 0.22	4.03 ± 0.19	4.24 ± 0.21	1 *	3.98 ± 0.21	4.10 ± 0.18	4.13 ± 0.17	4.16 ± 0.21	1*
HDL chol.	1.50 ± 0.06	1.43 ± 0.08	1.54 ± 0.05	1.51 ± 0.06	NS	1.50 ± 0.07	1.51 ± 0.06	1.57 ± 0.09	1.54 ± 0.08	NS
HDL_2 chol.	0.35 ± 0.04	0.25 ± 0.03	0.35 ± 0.04	0.46 ± 0.13	NS	0.36 ± 0.05	0.32 ± 0.06	0.36 ± 0.07	0.31 ± 0.05	NS
HDL_3 chol.	1.13 ± 0.06	1.16 ± 0.08	1.18 ± 0.04	0.99 ± 0.11	NS	1.13 ± 0.05	1.16 ± 0.04	1.26 ± 0.06	1.14 ± 0.07	NS
LDL chol.	1.83 ± 0.12	2.07 ± 0.15	2.26 ± 0.15	2.39 ± 0.15	1*	2.20 ± 0.18	2.31 ± 0.17	2.39 ± 0.12	2.38 ± 0.16	1*
Chol/HDL	2.38 ± 0.13	2.70 ± 0.21	2.65 ± 0.14	2.84 ± 0.15	1*	2.78 ± 0.20	2.82 ± 0.17	2.75 ± 0.18	2.75 ± 0.14	p<0.05 cycle 1 only
LDL/HDL	1.26 ± 0.10	1.51 ± 0.13	1.50 ± 0.13	1.62 ± 0.12	1	1.56 ± 0.16	1.63 ± 0.16	1.63 ± 0.16	1.62 ± 0.13	1
SHBG	48 ± 4	116 ± 15	143 ± 16	115 ± 12	1*	41 ± 3	78 ± 6	94 ± 9	85 ± 8	1*

No significant difference between groups pre Tx for any parameter.

1 All results significantly different from baseline, $p<0.05$. Change over study period significant, $p<0.05$.

* Time effect - Cycle 1 and 3 significantly different (3 and 6 similar).

NS No significant treatment or time effect or treatment-time interaction.

Analysis of variance showed statistically significant increases in total cholesterol, LDL-cholesterol and total triglycerides with time in both groups, but there were no statistically significant differences between the two treatment groups. Duncan's multiple range test showed that all three measures appeared to stabilise between cycles 3 and 6. HDL, HDL_2 and HDL_3-cholesterol subfractions were not significantly different from baseline at any time. Ratios of cholesterol/HDL-cholesterol and LDL-cholesterol/HDL-cholesterol increased significantly during cycle 1 of treatment in both groups, but stabilised thereafter and were maintained during cycles 3 and 6 (Table III).
There were no significant changes in fasting glucose or glycosylated haemoglobin (Table IV) and no woman had an increase of more than 9%.

Table IV Oral contraceptive effects on carbohydrate metabolism

Mean ± SEM

	SHD 415G				EE/LN			
	Pre RX	Cycle 1	Cycle 3	Cycle 6	Pre-RX	Cycle 1	Cycle 3	Cycle 6
Glucose (mmol/l)	5.2 ± 0.1	4.9 ± 0.1	4.9 ± 0.1	5.0 ± 0.1	4.9 ± 0.5	4.9 ± 0.1	4.9 ± 0.1	4.9 ± 0.1
Glycosylated Haemoglobin (%)	6.5 ± 0.12	6.8 ± 0.3	6.6 ± 0.2	6.5 ± 0.2	6.7 ± 0.7	6.9 ± 0.2	6.9 ± 0.2	6.8 ± 0.2

Values not significantly different.

Sex hormone binding globulin (SHBG) increased 2- to 3-fold.
Both OCs caused a significant increase in the prothrombotic factors - fibrinogen and factor VII. Fibrinogen increased during cycle 1 and stabilised thereafter, whilst factor VII activity increased between cycles 3 and 6. Antithrombin III, an important inhibitor of coagulation, fell significantly during cycle 1 and remained lower. For each time point, the results for women on the two different OCs were not significantly different (Table V).

Table V Effect of oral contraceptives on some coagulation factors

Mean percentage change from pretreatment ± SEM

	SHD 415G			EE/LN		
	Cycle 1	Cycle 6	Statistical Analysis	Cycle	Cycle 6	Statistical Analysis
Fibrinogen	28 ± 9	49 ± 12	Rise from baseline - $p < 0.05$ Time effect - NS	18 ± 6	15 ± 4	Rise from baseline - $p < 0.05$ Time effect - NS
Factor VII	12 ± 8	50 ± 13	Cycle 6 results sig.diff. from baseline Time effect - $p<0.05$	13 ± 12	22 ± 11	Cycle 6 results sig.diff. from baseline Time effect - $p<0.05$
Antithrombin III	-10 ± 4	-14 ± 6	Reduction from baseline $p < 0.05$ Time effect - NS	-9 ± 3	-16 ± 3	Reduction from baseline $p < 0.05$ Time effect - NS

The difference in changes with the two different treatments was not statistically significant (NS).

There were a total of 60 reported side effects: 11 women on SHD 415G and 16 on EE/LN reported one or more side effects. None were considered serious or unexpected for women on OCs. Those most frequently reported were nausea, headaches, breast tenderness and breakthrough bleeding (similar in each group, X^2 test).
Five patients (4 on SHD 415G) withdrew from the study. However, only 1 reported adverse symptoms (persistent headaches) and was withdrawn. The others failed to attend for follow-up visit

Discussion

Triphasic oral contraceptives were developed with the rationale that a reduction in the total steroid content would reduce side effects and metabolic effects whilst maintaining a high degree of contraceptive efficacy. This study compared two triphasics, one containing EE and LN, the other containing a new progestagen, gestodene. Assessment of weight, blood pressure and dietary changes showed that the two preparations had very little obvious physical effect, and the reported side effects were not significantly different.

Monitoring for 6 cycles with follow-up assessments during cycles 1, 3 and 6 showed no significant differences in serum lipoproteins between the SHD 415G and EE/LN treatment groups. Both OCs did, however, cause small but significant increases in total cholesterol, triglycerides, LDL-cholesterol and the ratio of total cholesterol /HDL-cholesterol and LDL-cholesterol / HDL-cholesterol. Changes in triglyceride and cholesterol concentration with OCs are well documented [25, 26], but most studies have not shown increased cholesterol levels with EE/LN, and only a few have shown a small increase in plasma triglycerides [25]. While oestrogens increase triglyceride levels, levonorgestrel has anti-oestrogenic actions which tend to reduce plasma triglycerides. The greater rise in triglycerides in women on SHD 415G thus appears to reflect a greater net oestrogen effect. The significance of a slight rise in triglycerides is unclear, although it could be important in people with a low HDL-cholesterol [5].
Changes in lipoprotein fractions are likely to be more important than total cholesterol or triglycerides in predicting a possible increased risk of CHD. LDL-cholesterol is positively associated with CHD risk and LDL-cholesterol was significantly increased by both preparations. HDL-cholesterol concentration, particularly that of HDL_2, is inversely related to CHD risk [3,4]. There were no significant changes in HDL-, HDL_2- or HDL_3-cholesterol subfractions, which is encouraging as gonane progestagens with anti-oestrogenic activity tend to decrease HDL-cholesterol [27], and some studies [28, 29] of EE/LN have suggested

a decrease in HDL_2-cholesterol linked to the anti-oestrogenic potential of LN and stimulation of hepatic lipase activity [30]. Oestrogens may counteract this effect [31] and this may explain why the triphasics did not reduce HDL levels. Our results concur with those of Larsson-Cohn [32], who found no significant difference in HDL-cholesterol with a triphasic EE 30/40/30 preparation with EE to LN ratios of 0.09 to 0.35 during the pill cycle. However, HDL-cholesterol/total cholesterol remained constant, whereas in our study total cholesterol/HDL-cholesterol and LDL-cholesterol/HDL-cholesterol ratios increased significantly. This ratio was also unchanged at 6 months in the study by Lepot et al. of both triphasics, but at 12 months the LDL/HDL cholesterol ratio was reduced with the gestodene preparation [33].
In general, the women had relatively low total and LDL-cholesterol concentrations at baseline and on OCs their levels were lower than men of a similar age. Relative cardiovascular risk does, however, appear to increase from a plasma cholesterol as low as 4.5 mmol/l [34] and it is theoretically possible that if the changes observed on the two OCs were maintained over a prolonged period, this could increase the women's CHD risk.
Coagulation factors VII and fibrinogen increased significantly in both groups. Fibrinogen levels increased from baseline after cycle 1 and then stabilised, but levels of Factor VII, which are also a sensitive indicator of change in the coagulation pathway, were significantly increased after cycles 3 and 6, suggesting that some time is required for the changes. Antithrombin III, an important inhibitor of coagulation, decreased significantly from baseline. A decrease has been observed with low dosage oestrogen combined monophasic preparations, but previous workers have reported normal [35] or even increased antithrombin III activity [36] associated with the use of triphasic OCs. These and the effects on other coagulation factors are discussed in more detail in a previous paper [37]. Overall, the findings are in agreement with other studies of OCs [38,11,39,40], which suggest a prothrombotic change; but all group means remained within the laboratory reference range. We did not, however, examine the fibrinolytic side of the coagulation cascade and cannot exclude an increase in fibrinolysis which could have redressed the balance. The relative importance of the findings is unclear, but an increase in fibrinogen and factor VII may be important as Meade et al. found a relationship between fibrinogen levels and factor VII c and CHD[41]. Mammen [38] has,however, suggested that circulating clotting factors may exceed levels needed to produce a thrombotic event in some normal people, so raised levels alone may not be an important factor. Sex hormone binding globulin (SHBG) concentration rises with oestrogen therapy [42] and it was monitored in this study to facilitate comparison of oestogenic potency of the two triphasic preparations, and the ability of the progestagen to suppress the oestrogenic effect. Both

preparations significantly increased SHBG levels, by 2-to 3-fold, the increase being greater with SHD 415G. Increases in SHBG are well documented with combined OCs [43]. The greater increase with SHD 415G was not surprising as LN has previously been shown to reduce the oestrogen-induced rise in SHBG[42]. It is interesting that with respect to the effect on plasma cholesterol and triglyceride levels, SHD 415G seemed to have a greater oestrogenic effect. Yet, the absence of effect on HDL levels was not that expected for a preparation with greater oestrogenic effect, unless perhaps the gestodene itself had a specific counteracting effect on HDL. Some higher oestrogen OCs [44] and gonane progestagens tend to reduce glucose tolerance. Potentiation by oestrogen has been suggested [45], but women on LN alone can have higher plasma glucose levels [46]. Neither of the triphasics that we studied affected fasting glucose, or glycosylated haemoglobin (which was measured as an index of longer term prevailing glucose levels). These parameters are, however, rather insensitive measures, and a more rigorous analysis of carbohydrate effects could have been performed using an intravenous glucose tolerance test and the minimal-modeling technique of Bergman.
This study indicates that despite the theoretical advantages, the triphasic containing gestodene is not obviously superior to that containing levonorgestrel. It must be noted, however, that the metabolic effects of both of the triphasics are much less than those of many older combined OCs[12]. The implications of the slight increase in lipoproteins levels and the moderate changes in coagulation factors, both potential risk factors for cardiovascular disease, remain uncertain. Yet it would seem appropriate to aim for preparations which have the minimum of metabolic effects.

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